Stylistic Drift as a Model Fingerprint: Measuring LLM Stylistic Signatures and Their Prompt Sensitivity in French Literary Rewrites

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Abstract

We study stylistic drift in LLM-rewritten French literary texts using function-word frequency vectors. For each original–rewrite pair, we define *shift* as the cosine distance between their L2-normalised 41-dimensional style vectors, and compare four commercial models (GPT-4o, Claude Sonnet 4.6, Mistral Small, Gemini 2.5 Flash) across 80 texts from Zola and Maupassant and two rewriting prompts. Three findings emerge. First, Gemini produces significantly larger stylistic drift than the other models ($\Delta = 0.23$, $p < 0.001$ after Bonferroni correction); GPT-4 and Mistral are statistically indistinguishable ($p = 1.0$). Second, surface structure is a stronger discriminative signal than function-word shift for model attribution: 15 surface features (sentence length, type-token ratio, punctuation density) yield 56.2% leave-one-out (LOO) accuracy on a 4-class problem (baseline 25%), outperforming 41-dimensional shift vectors (40.6%); combining both representations reaches 57.5%, $2.3\times$ the random baseline. Third, inter-prompt robustness is weak across all models: per-text Pearson correlations between P1 and P2 shift series are non-significant for Gemini and Claude ($r < 0.07$, $p > 0.6$) and borderline significant for GPT-4 and Mistral ($r \approx 0.28$, bootstrap 95% CI barely excluding zero). Model rankings are preserved at the aggregate level, but individual-text shift is not predictable across prompts, suggesting that stylistic fingerprints reflect distributional tendencies of models rather than stable per-document properties. Code and data: `https://github.com/riadmaouchi/llm-style-fingerprints` (DOI: 10.5281/zenodo.20402754).

1 Introduction

Large language models rewrite text fluently. But do they rewrite *neutrally*? Prior work on authorship attribution has long established that function words—articles, pronouns, prepositions, conjunctions—carry stylistic information that is largely unconscious and difficult to suppress [Mosteller and Wallace, 1964, Burrows, 1987, Juola, 2015]. If LLMs systematically alter the distribution of function words when paraphrasing, they leave a measurable stylistic imprint on the output.

The dominant framing in recent work is *detection*: given a text, decide whether it was written or assisted by an LLM [Uchendu et al., 2023, Sadasivan et al., 2023, Liang et al., 2024]. We adopt a different perspective. Rather than asking "is this AI-generated?", we ask: *Is the stylistic fingerprint of an LLM robust to the instruction it receives?*

This question matters for two reasons. Practically, if the signature is instruction-dependent, detection methods trained on one prompt will generalise poorly. Theoretically, it bears on what a "style" actually is for an LLM: a stable latent property of the model, or an emergent pattern driven by the interaction between model and prompt.

We operationalise stylistic drift as the cosine distance between the function-word frequency vectors of an original text and its rewrite (§3), and compare four commercial models across 80 French literary excerpts and two prompts. Our main contributions are:

1. A reproducible framework for measuring LLM stylistic drift on a French literary corpus (40 Zola + 40 Maupassant excerpts, $\sim$120 words each), with all API outputs archived.
2. Evidence that model rankings are preserved across prompts but per-text shift correlations are weak or null for all four models (bootstrap CIs include zero for three out of four), suggesting that stylistic fingerprints are distributional descriptors of models rather than stable per-document properties.
3. A classification analysis across eight feature configurations showing that surface statistics (sentence rhythm, type-token ratio, punctuation patterns) outperform function-word shift vectors for 4-class model attribution (56.2% vs. 40.6% LOO accuracy); the combined 16-feature representation reaches 57.5%, 2.3$\times$ the random baseline. The primary confusion is structural (GPT-4 $\leftrightarrow$ Mistral), consistent with their statistically indistinguishable shift magnitudes.

2 Related Work

Authorship attribution and function words. The use of function-word frequencies for authorship attribution dates to Mosteller and Wallace [1964] and was formalised by Burrows [1987]'s Delta method, which measures cosine distance between z-scored frequency vectors. Our shift measure is structurally related to Delta, with two differences: (i) we compare an original to its rewrite rather than two authors, making the baseline text-specific; and (ii) we use the raw shift vector $v(t') - v(t)$ as a feature for classification, rather than distance alone. Studies in authorship attribution routinely achieve 80–90%+ accuracy [Stamatatos, 2009, Koppel et al., 2009], but on corpora of thousands of samples per author exhibiting stable, lifelong stylistic patterns. Our setting is structurally harder: LLMs are not authors but instruction-following systems, and our corpus contains only 80 texts. The comparison is informative but not direct.

LLM-generated text detection. Uchendu et al. [2023] benchmark multiple detectors across generative models and report classification accuracy above 80% in a controlled setting. However, their task differs from ours in two critical ways: they classify full-length generated articles (not short rewrites), and they distinguish generative models from each other or from humans (not from the same human text rewritten under different instructions). Both factors make their task easier, and their results should not be taken as an upper bound for our problem. Sadasivan et al. [2023] prove that detection accuracy degrades fundamentally under paraphrasing, which is consistent with our 40.6% accuracy for shift vectors alone: rewrites are by design closer to the original than freely generated text. Liang et al. [2024] demonstrate that surface-level stylometric detectors produce false positives for non-native speakers, a bias our function-word approach may share when applied to contemporary text.

Prompt sensitivity and style robustness. Guo et al. [2023] compare ChatGPT outputs to human experts but hold the prompt fixed; prompt sensitivity is not studied. Zhu et al. [2023] show that casual paraphrasing defeats detectors, implying sensitivity to instruction, but do not measure per-text correlation of shift across prompt variants. Style transfer evaluation [Mir et al., 2019, Tikhonov and Yamshchikov, 2019] uses function-word features to assess how much style is preserved after transfer, a complementary perspective to our shift measurement. To our knowledge, no prior work directly measures whether the stylistic shift produced by a given LLM on a given text is consistent across semantically distinct rewriting instructions. This is the gap our robustness analysis (§4.3) addresses, albeit at modest corpus scale.

3 Method

3.1 Corpus

We use 80 French literary excerpts: 40 from Zola (*Germinal*, *L'Assommoir*, *Nana*, *La Bête humaine*, *La Débâcle*, and others) and 40 from Maupassant (*Boule de suif*, *La Parure*, *Yvette*, *Miss Harriet*, and others).

Each excerpt contains approximately 120 words. All texts are in the public domain.

3.2 Rewriting Prompts

Each excerpt is rewritten by four commercial LLMs under two prompts:

- **P1** (neutralisation): *"Réécris ce texte dans un style neutre et factuel, en conservant le sens."*
- **P2** (simplification): *"Reformule ce texte en simplifiant le vocabulaire, pour le rendre accessible au grand public."*

The four models are GPT-4o, Claude Sonnet 4.6, Mistral Small (`mistral-small-latest`), and Gemini 2.5 Flash. All calls use temperature 0.7 and a maximum of 1 500 output tokens. The 640 rewrites (80 texts $\times$ 2 prompts $\times$ 4 models) are archived in the repository.

3.3 Style Vectors and Shift

Following the function-word tradition [Burrows, 1987, Stamatatos, 2009], we represent each text as a 41-dimensional vector of relative function-word frequencies (articles, pronouns, prepositions, conjunctions, and a curated set of discourse markers associated with formal register: *tandis, pourtant, néanmoins, notamment, afin*). Vectors are L2-normalised.

The *stylistic shift* of a rewrite t' with respect to its original t is:

$$\delta(t, t') = 1 - \frac{v(t) \cdot v(t')}{\|v(t)\| \, \|v(t')\|} \tag{1}$$

where $v(\cdot)$ denotes the style vector. A shift of 0 indicates identical function-word profiles; a shift of 1 indicates maximal divergence.

We also define a *shift vector* $\mathbf{s} = v(t') - v(t)$, which captures the *direction* of the stylistic change in addition to its magnitude. Shift vectors are used as features for the LDA projection and classifier.

3.4 Surface Features

Beyond function-word shift, we extract two families of surface features.

Surface basic (5 features): mean sentence length, sentence length standard deviation, type-token ratio, punctuation density, and word count.

Surface extended (15 features): the five basic features plus comma rate, semicolon rate, average word length, ellipsis density, em-dash density, French guillemet density, parenthesis rate, letter density, long-word ratio ($>$8 characters), and short-sentence ratio ($<$5 words per sentence).

We also define a *scalar shift* feature: the single cosine distance $\delta(t, t')$ between original and rewrite, which encodes the intensity of the stylistic transformation independently of its direction.

3.5 Statistical Tests

Pairwise comparisons use permutation tests (10 000 permutations, two-sided mean-difference statistic) with Bonferroni correction across all $\binom{4}{2} = 6$ pairs. Bootstrap confidence intervals (95%) are computed with $n = 5\,000$ resamples. Inter-prompt robustness is measured by Pearson r between the per-text shift series under P1 and P2 for each model.

Model	Mean shift	95% CI
Gemini 2.5 Flash	0.230	[0.204, 0.256]
Claude Sonnet 4.6	0.170	[0.147, 0.195]
Mistral Small	0.139	[0.124, 0.157]
GPT-4o	0.132	[0.113, 0.152]

Table 1: Mean cosine shift (P1) and 95% bootstrap CI ($n = 80$).

4 Results

4.1 Stylistic Drift under P1

Table 1 reports mean shift and bootstrap confidence intervals for each model under P1. Gemini produces the largest drift by a substantial margin ($\Delta = 0.230$), while GPT-4 and Mistral are nearly identical ($\Delta = 0.132$ and 0.139 respectively).

Pairwise permutation tests (Table 2) show that Gemini is significantly different from all other models ($p < 0.01$ in all three comparisons), while GPT-4 and Mistral are indistinguishable ($p = 1.0$) and Claude occupies an intermediate, non-significant position.

Pair		p_{corr}	Sig.
GPT-4	vs Gemini	< 0.001	✓
Mistral	vs Gemini	< 0.001	✓
Claude	vs Gemini	0.007	✓
GPT-4	vs Claude	0.103	
Claude	vs Mistral	0.262	
GPT-4	vs Mistral	1.000	

Table 2: Pairwise permutation tests (10 000 permutations), Bonferroni-corrected ($k = 6$).

4.2 Feature Analysis and Classification

We project shift vectors onto the two leading Linear Discriminant Analysis (LDA) components. The LDA projection reveals that GPT-4 and Mistral share an almost identical centroid, while Gemini occupies a clearly separated region. Claude is intermediate, with substantial overlap with both clusters.

Table 3 reports LOO logistic regression accuracy for eight feature configurations on the 4-class attribution problem ($n = 320$, 80 texts $\times$ 4 models).

Feature set	Dim	Accuracy
Surface ext. + scalar shift $\star$	16	**57.5%**
Surface extended (15 features)	15	56.2%
Shift + surface basic (combined)	46	43.8%
Surface statistics basic (5 feat.)	5	41.9%
Shift vectors (function words)	41	40.6%
Raw rewrite FW vectors	41	39.4%
Char n-grams TF-IDF (3–6)	5000	33.1%
Original FW vectors (sanity)	41	0.0%
Random baseline	—	25.0%

Table 3: LOO logistic regression accuracy for 4-class model attribution ($n = 320$, LogReg $C = 3$). FW = function-word; $\star$ = best configuration. Surface statistics outperform function-word shift vectors at every dimensionality tested.

Four observations follow.

Surface structure dominates function-word space. Extended surface features alone (15 measures: mean and standard deviation of sentence length, type-token ratio, punctuation density, comma and semicolon rates, average word length, ellipsis and em-dash density, French guillemet density, parenthesis rate, letter density, long-word ratio, short-sentence ratio) reach 56.2% LOO accuracy, a 15.6-point gap over 41-dimensional shift vectors (40.6%). This is the primary finding: the structural footprint of each model is more discriminative than its function-word signature.

Scalar shift adds complementary signal. Adding a single feature—the cosine distance $\delta(t, t')$ between original and rewrite—to the 15 surface features lifts accuracy to 57.5%, 2.3× the 25% random baseline. The scalar shift encodes the intensity of stylistic transformation, a signal that surface features do not capture.

Shift subtraction provides marginal gain alone. Shift vectors (40.6%) outperform raw rewrite function-word vectors (39.4%) by only 1.3 percentage points, limiting the methodological claim for the subtraction step when used as the sole feature family.

High-dimensional methods overfit. Char n-grams (5 000 features, 33.1%) underperform despite high dimensionality, consistent with overfitting in LOO on $n = 320$.

The original-only condition reaches 0%, confirming that the classification signal comes entirely from the rewrite, not the source text. The primary confusion in all conditions is between GPT-4 and Mistral, consistent with their indistinguishable mean shift ($p = 1.0$).

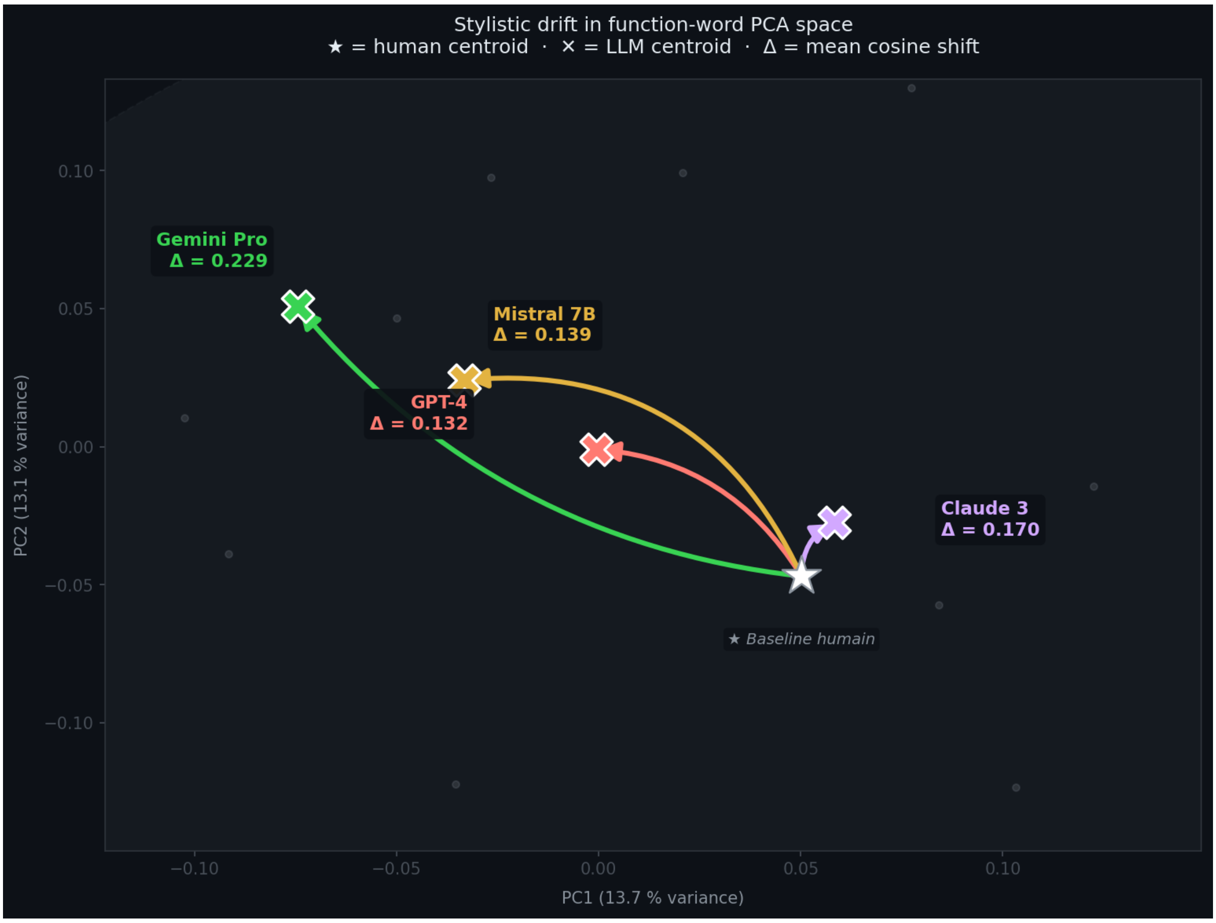

Figure 1: Drift trajectories in PCA function-word space. Star (⋆): human centroid. Crosses (×): LLM centroids. Arrows show direction and magnitude of stylistic displacement from the human baseline. Δ = mean cosine shift.

4.3 Inter-Prompt Robustness

Table 4 reports, for each model, the Pearson correlation between per-text shift under P1 and P2, together with the mean shift under each prompt.

Model	r	**95% CI**	p	$\bar{\delta}_{P1}$	$\bar{\delta}_{P2}$
GPT-4	0.28	[0.04, 0.51]	0.011	0.132	0.096
Mistral	0.28	[−0.00, 0.51]	0.011	0.139	0.130
Gemini	0.06	[−0.17, 0.30]	0.602	0.231	0.191
Claude	0.03	[−0.21, 0.32]	0.778	0.170	0.176

Table 4: Inter-prompt robustness: Pearson r between per-text shifts under P1 and P2 with bootstrap 95% CI ($n = 80$, 5 000 resamples).

Per-text correlations are weak or null across all models. Only GPT-4 reaches nominal significance ($p = 0.011$), but its bootstrap CI is wide [0.04, 0.51], indicating substantial uncertainty. Mistral's lower CI bound is essentially zero (-0.00). Gemini and Claude show no significant correlation ($p > 0.6$). P2 consistently produces lower mean shift than P1 for GPT-4 and Mistral (simplification is less stylistically disruptive than neutralisation), while Claude's mean shift is virtually identical under both prompts (0.170 vs. 0.176) despite the null correlation.

5 Discussion

Aggregate ranking vs. individual prediction. All four models preserve the same relative ranking under P1 and P2 (Gemini $>$ Claude $>$ Mistral $\approx$ GPT-4). This aggregate robustness is the strongest claim this study can make: the choice of rewriting prompt does not change *which* model shifts style the most.

However, individual-level inference is not supported. With $n = 80$ and wide bootstrap CIs, only GPT-4 nominally crosses the significance threshold, and even that result ($r = 0.28$, CI [0.04, 0.51]) should be treated cautiously. Knowing that a given text was strongly shifted under P1 gives essentially no reliable prediction for its shift under P2 for any of the four models.

On classification accuracy and feature choice. The eight feature configurations reveal a clear hierarchy: surface statistics (sentence rhythm, lexical richness, punctuation patterns) consistently outperform function-word shift vectors at every dimensionality tested. Extended surface features alone reach 56.2%—a 15.6-point gap over shift vectors (40.6%)—with the combined 16-feature set peaking at 57.5%. This is the primary empirical finding: the structural footprint of each LLM (how it segments sentences, how it varies vocabulary, how it uses punctuation) is more discriminative than its function-word signature. Shift vectors provide only a 1.3 percentage-point improvement over raw rewrite vectors, which limits the methodological claim for the subtraction step when used as the sole feature family. The gap between our best result (57.5%) and the 80%+ reported by Uchendu et al. [2023] is explained by task design: full-length generation tasks offer far more signal than short rewrites of existing texts, and our 4-class problem includes a structurally hard pair (GPT-4 $\leftrightarrow$ Mistral, $p = 1.0$) that suppresses overall accuracy.

Limitations. $n = 80$ excerpts from two 19th-century French authors limits generalisability and statistical power. Bonferroni correction on 6 pairwise tests is conservative; some differences that are real may not reach significance here. The function-word list includes markers selected partly for their known association with LLM formal register (*tandis, néanmoins, notamment*), introducing a mild bias toward models that produce formal connectives. We have no human rewriter baseline: Koppel et al. [2009] show that even human authors exhibit significant intra-author variance across genres; a human asked to neutralise then simplify the same text might produce similarly low per-text shift correlations, in which case our null results would reflect task difficulty rather than a model-specific property. Finally, all models were queried at a fixed snapshot in time; model updates may alter these signatures.

6 Conclusion

We measured stylistic drift in 640 LLM rewrites of French literary texts using cosine distance in 41-dimensional function-word space. Gemini produces the largest drift and is the only model clearly distinguishable from all others; GPT-4 and Mistral are statistically indistinguishable ($p = 1.0$). At the aggregate level, model rankings are stable across two prompts. A systematic feature comparison across eight configurations reveals that surface statistics (sentence rhythm, lexical richness, punctuation patterns) outperform function-word shift vectors for model attribution: 56.2% vs. 40.6% LOO accuracy on 4 classes; combined features reach 57.5%, 2.3$\times$ the 25% random baseline. The dominant attribution error is structural (GPT-4 $\leftrightarrow$ Mistral), consistent with their indistinguishable shift magnitudes. At the individual-text level, per-text shift correlations across prompts are weak or null for all four models (bootstrap CIs include or nearly include zero). We conclude that LLM stylistic fingerprints are reliable comparative descriptors at the aggregate level, that surface structure is the more discriminative signal for attribution, and that individual-text prediction remains limited by corpus size and the structural similarity of some model pairs.

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All API outputs, code and data used in this study are archived at `https://github.com/riadmaouchi/llm-style-fingerprints` (DOI: 10.5281/zenodo.20402754). No human subjects were involved.

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